

Executive Summary

A MERN-stack prototype – “**Smart Rural Healthcare Appointment & Follow-up Management System**” – will streamline Primary Health Centre (PHC) workflows by digitising patient registration, appointment booking, consultation recording, immunisation tracking and automated reminders. The system will follow Tamil Nadu (TN) health guidelines and Ayushman Bharat Digital Mission (ABDM) standards: for example, linking patient records via ABHA (health ID) and “Scan & Share” QR registration to cut wait times [14†L52-L60] [14†L128-L136] . Built on TN’s secure intranet (TNSWAN) [12†L142-L150] , it uses JWT auth and HTTPS for data privacy.

UI Pages (with Purpose & Role)

Page	Purpose	Primary Role(s)
Login / Register	Authenticate users (email/ABHA, password).	All (Patient, Staff, Admin)
Patient Dashboard	View upcoming appointments, reminders, immunisation status.	Patient
Appointment Booking	Book/reschedule/cancel PHC appointments.	Patient, Receptionist
Medical History	View past consultations and prescriptions.	Patient, Doctor
Consultation Page	Enter diagnosis, prescription and schedule follow-ups.	Doctor
Immunisation Tracker	Track maternal/child vaccine schedule and due dates.	Patient, ANM/VHN/ Staff
Notifications Center	Show appointment reminders and alerts.	All (Patient, Staff)
Doctor Dashboard	List today’s appointments and pending follow-ups.	Doctor
Reception Dashboard	Overview of today’s patients and queue management.	Receptionist/Staff
Admin Dashboard	Hospital stats, manage users (doctors/staff), reports.	Admin/Manager
User/Profile Settings	Update profile, change password, link ABHA, etc.	All

Modules (Key Features & Priority)

Module	Key Features	Priority
Authentication	Secure JWT login/registration; role-based access; password reset.	MVP

Module	Key Features	Priority
Patient Management	Store patient demographics (ABHA, contact, address, maternal ID); search/edit profiles.	MVP
Appointment System	Book/select doctor, date/time; view calendar; cancel/reschedule.	MVP
Consultation/EMR	Record doctor's notes, diagnosis, prescriptions, and schedule follow-up.	MVP
Follow-up & Reminder	Auto-schedule reminders (SMS/WhatsApp/Email) for next visits or meds; track status.	MVP
Immunisation Tracker	Manage maternal/child vaccine schedule; flag due/overdue shots; send reminders.	MVP
Notifications	Send/display alerts (appointments due, follow-ups, vaccination alerts).	MVP
Dashboard & Reporting	Patient/appointment stats, alerts (doctor: today's list; admin: hospital metrics).	MVP
User & Role Management	Admin can add/doctors, set roles/permissions; staff accounts.	MVP
Integrations (Govt)	ABDM Scan&Share QR registration; link ABHA records 【14†L52-L60】 【14†L128-L136】 .	Future (Demo)
Interoperability	(Mock) Share patient history across hospitals/PHCs via secure API/TNSWAN 【12†L142-L150】 .	Future
Telemedicine/ Chatbot	(Optional) Virtual consults; AI chat reminders.	Future
Analytics (Optional)	(Optional) Trends on footfall, common illnesses, vaccination coverage.	Future

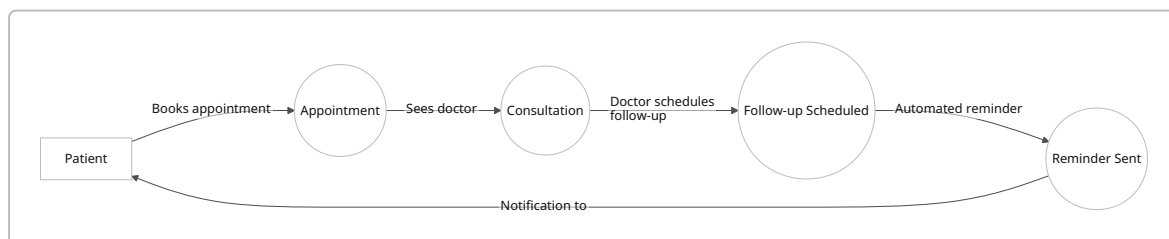
MongoDB Collections (Name & Key Fields)

Collection	Key Fields
Users	_id, name, email, password_hash, role (Patient/Doctor/Staff/Admin), hospital_id, ABHA_id, contact, status
Patients	_id, user_id (ref Users), DOB, gender, address, maternal_ID, emergency_contact, ABHA_id, etc.
Appointments	_id, patient_id, doctor_id, hospital_id, datetime, reason, status (booked/completed/missed).
MedicalRecords	_id, patient_id, doctor_id, date, notes, prescriptions, followup_days, attachments (optional).
Immunizations	_id, patient_id, vaccine, due_date, status (pending/completed), administered_date.

Collection	Key Fields
Notifications	<code>_id</code> , user_id, type (appointment/followup/immunization), message, sent_at, read_flag.
Hospitals	<code>_id</code> , name, address, type (PHC/UPHC/CHC), doctors_list, TNSDC_endpoint (for integration).

Third-Party Services and Security Notes

- **Messaging:** Integrate SMS/WhatsApp via providers (e.g. Twilio, TextLocal) and SMTP email (SendGrid/Mailgun) for sending reminders/alerts.
- **Hosting:** Deploy on AWS/GCP or DigitalOcean with HTTPS. In TN context, optionally host on state data center (TNSWAN) for secure intranet access 【12†L142-L150】 .
- **Authentication:** Use JWT tokens, secure password hashing (bcrypt) and HTTPS to protect PHI. Enforce strong role-based access (admins, staff, doctors, patients).
- **Data Privacy:** Follow NDHM/ABDM privacy rules – patient consent for sharing records, data encryption at rest/in transit. Log access for audit.
- **Compliance:** Design schemas to allow linking with ABHA IDs 【14†L52-L60】 ; ensure interoperability with ABDM APIs.



Sources: Tamil Nadu eGov (TNSWAN network) 【12†L142-L150】 , Ayushman Bharat Digital Mission guidelines 【14†L52-L60】 【14†L128-L136】 , and MERN architecture best practices 【18†L85-L93】 .